Single-Phase Power Transformer Sizing

A Practical Methodology for High-Frequency, High-Power Designs

1 Scope

This document summarizes the engineering steps required to dimension a single-phase power transformer for switch-mode converter applications (e.g. PSFB, DAB, isolated DC/DC stages), covering:

- turns ratio determination from the voltage transfer function,
- core material selection as a function of operating frequency,
- core sizing via the area-product method,
- winding design considerations (skin/proximity effects, litz wire vs. foil),
- a worked numerical example at 800 V/800 V, 10 kHz, 250 kW.

2 Turns Ratio

For an ideal isolated transformer driven by a quasi-square or square voltage waveform (as typical of PSFB, DAB, or any hard/soft-switched bridge), the volt-second balance on each winding gives Faraday's law in its waveform-corrected form:

$$V = K_f \, N \, A_c \, B_m \, f \tag{1}$$

where:

- V = RMS (or peak, depending on K_f convention) voltage applied to the winding [V],
- N = number of turns,
- A_c = core cross-sectional area [m^2],
- B_m = peak flux density [T],
- f = switching/operating frequency [Hz],
- K_f = waveform factor: $K_f = 4$ for a symmetric square wave, $K_f = 4.44$ for a pure sinusoid, intermediate values for trapezoidal/quasi-square waveforms with duty cycle $D < 0.5$ (e.g. $K_f \approx 4D$ for a unipolar square wave of duty D).

The turns ratio is fixed by the required voltage transformation:

$$n = \frac{N_1}{N_2} = \frac{V_1}{V_2} \tag{2}$$

For converters requiring voltage step change under varying duty cycle (e.g. PSFB with output inductor), n is chosen at the nominal/maximum duty point, with margin for line and load regulation.

3 Core Material Selection vs. Frequency

The dominant constraint at a given frequency is core loss, which is approximately:

$$P_{core} \approx k_c \, f^{\alpha} \, B_m^{\beta} \, V_{core} \tag{3}$$

(Steinmetz equation), where α, β and k_c are material-dependent. The choice of material is therefore frequency-driven:

Material	Typical f range	B_{sat} [T]	Notes
Grain-oriented Si-Fe (laminated)	DC–1 kHz	1.6–2.0	High saturation, but eddy-current losses scale with f^2t^2; lamination thickness limits practical use above ∼1 kHz.
Amorphous (Fe-based)	1–10 kHz	1.5–1.6	Lower loss than Si-Fe at the same f, good compromise for medium-frequency, high-power transformers.
Nanocrystalline (e.g. Vitroperm, Finemet)	1–150 kHz	1.2–1.5	High B_{sat} retained at MF, low specific losses; preferred for high-power MF transformers (DAB, solid-state transformers, traction).
MnZn ferrite	20 kHz–500 kHz	0.35–0.5	Low B_{sat} limits power density at low kHz; becomes attractive above ∼50 kHz where its losses remain low and nanocrystalline/amorphous losses rise.
NiZn ferrite	>500 kHz	0.3–0.4	High-frequency, low-power applications; not relevant for high-power MF transformers.

Rule of thumb: for a given frequency, select the material that allows the highest usable B_m (limited by core loss density, typically targeting 0.5 W/cm^3 to 1.5 W/cm^3 depending on cooling) while satisfying thermal constraints. At 10 kHz and high power, nanocrystalline or amorphous tape-wound cores typically outperform ferrite because their higher B_{sat} allows a smaller, lighter core for the same volt-seconds, even though their specific core losses per unit volume at 10 kHz are higher than ferrite's — the reduction in required volume usually wins for power levels in the hundreds of kW.

4 Core Sizing: Area-Product Method

The area product $A_p = A_c \cdot A_w$ (core area times window area) relates the apparent power transferred to the achievable current density and flux density, and is the standard first-pass sizing tool (McLyman):

$$A_p \, [\mathrm{cm}^4] = \frac{P_t \cdot 10^4}{K_f \, K_u \, B_m \, f \, J} \tag{4}$$

where:

- P_t = total apparent power handled by the transformer, $P_t \approx P_{in} + P_{out} \approx 2P$ for an isolated transformer carrying power P on each side,

- K_u = window utilization factor (≈ 0.2–0.3 for round/litz wire with insulation and creepage, up to ≈ 0.4–0.5 for foil/busbar windings),
- B_m = peak flux density [T] (material- and loss-limited, see above),
- f = frequency [Hz],
- J = RMS current density [A/cm^2] (cooling-dependent: $\sim$200–350 for natural convection, up to 500+ with forced air/liquid cooling).

Once A_p is known, a core (or stack of cores) is selected from a manufacturer catalogue with matching or slightly larger A_p, giving A_c and A_w directly. Equation (1) then yields the turns count for each winding.

5 Winding Design

At MF/HF, conduction losses are dominated by skin and proximity effects rather than DC resistance:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0 \mu_r}} \tag{5}$$

For copper, $\delta \approx 7.5/\sqrt{f\,[\text{Hz}]}$ cm. At 10 kHz, $\delta \approx 0.66$ mm, meaning conductor thickness should not exceed roughly $2\delta \approx 1.3$ mm in any single layer.

- **Litz wire**: appropriate for moderate current windings where many strands keep each strand below δ; strand count and bundle geometry must be checked for proximity losses in multi-layer windings.
- **Copper foil**: preferred at very high current, low turn-count windings (as in the example below); foil thickness $\leq 2\delta$, interleaved with thin insulation, often the only practical option above a few hundred amperes RMS per turn.
- **Interleaving**: primary/secondary interleaving reduces the MMF excursion across the window and is essential to control proximity losses at MF in high-current designs.

6 Leakage Inductance Estimation

Leakage inductance arises from the imperfect coupling between primary and secondary windings and is set almost entirely by winding geometry rather than core material. For concentric (or foil) windings arranged radially in the window, the standard analytical estimate (Erickson & Maksimovic; McLyman)[2, 1] referred to the primary is:

$$L_{lk} \approx \frac{\mu_0 N_1^2 \,\text{MLT}}{h_w} \left[\frac{b_1 + b_2}{3} + \delta\right] \tag{6}$$

where:

- MLT = mean length per turn [m],
- h_w = winding (window) height available for the turns [m],
- b_1, b_2 = total radial (or axial, depending on construction) build-up of the primary and secondary windings respectively [m],
- δ = thickness of the inter-winding insulation/isolation gap [m].

Equation (6) shows the two key levers for leakage reduction:

- **Increasing** h_w (taller, narrower window) spreads the same turns over a larger MMF-cancellation path, reducing L_{lk}.
- **Interleaving** the primary and secondary into m alternating sections divides the effective leakage by approximately m^2:

$$L_{lk,\,interleaved} \approx \frac{L_{lk}}{m^2} \tag{7}$$

For DAB or PSFB topologies, leakage inductance is often a wanted design parameter (it forms part or all of the power-transfer/resonant inductance), so this calculation should be checked against the value required by the converter control law, not only minimized.

7 Loss Estimation

Total transformer loss is the sum of core loss and winding (copper) loss.

Core Loss

Core loss density is obtained from the manufacturer's Steinmetz-type curves at the operating f and B_m (datasheet value, in W/kg or W/cm^3), then multiplied by the effective core volume/mass:

$$P_{core} = P_v(f, B_m) \cdot V_e \tag{8}$$

where $V_e \approx A_c \cdot l_e$, with l_e the mean magnetic path length (approximately the perimeter around the window, $l_e \approx 2(\sqrt{A_c} + \sqrt{A_w})$ for a first-pass square-leg estimate).

Winding (AC) Loss

DC loss per winding is $P_{dc} = I^2 R_{dc}$, with $R_{dc} = \rho\, l / A_{Cu}$ and $l = N \cdot \mathrm{MLT}$. At MF, the actual loss is higher due to skin and proximity effects, captured by the AC resistance factor $F_R = R_{ac}/R_{dc}$. For a foil/layer winding, Dowell's equation[8] gives:

$$F_R = \Delta \left[\frac{\sinh 2\Delta + \sin 2\Delta}{\cosh 2\Delta - \cos 2\Delta} \right] + \frac{2(p^2 - 1)}{3}\, \Delta \left[\frac{\sinh \Delta - \sin \Delta}{\cosh \Delta + \cos \Delta} \right] \tag{9}$$

where $\Delta = t/\delta$ (conductor thickness over skin depth) and p is the number of layers between successive interleaving nulls (i.e. $p = N$ for a non-interleaved winding, reduced roughly to N/m for m-section interleaving). The second (proximity) term dominates rapidly as p grows — this is the quantitative justification for the interleaving recommendation in the winding design section.

8 Insulation and Isolation

Creepage and clearance distances are dictated by the working voltage, altitude, pollution degree, and required isolation class (basic/supplementary/reinforced) per IEC 60664-1 or the relevant converter standard. Tape-wound cores with bobbin or potted construction are common; for busbar/foil windings, solid insulation (e.g. polyimide, Nomex, epoxy-glass) must be verified for partial-discharge-free operation at the working dV/dt, which can be severe with hard-switched square waves.

9 Worked Example: 800 V/800 V, 10 kHz, 250 kW

Assumptions

- Square-wave excitation (PSFB/DAB-type bridge), $K_f = 4$.
- Turns ratio $n = 1$ (800 V/800 V).
- Nanocrystalline tape-wound core, $B_m = 1.0\,\mathrm{T}$ (derated from $B_{sat} \approx 1.2\,\mathrm{T}$ for MF loss margin).
- $K_u = 0.3$ (foil winding, generous insulation/creepage at this power level).
- $J = 300\,\mathrm{A/cm^2}$ (forced-air or liquid-cooled busbar/foil winding).

Step 1 — Apparent power

$$P_t = 2P = 2 \times 250\,\mathrm{kW} = 500\,\mathrm{kW} \tag{10}$$

Step 2 — Area product

$$A_p = \frac{500{,}000 \times 10^4}{4 \times 0.3 \times 1.0 \times 10{,}000 \times 300} \approx 1389\,\mathrm{cm^4} \tag{11}$$

This is a large core (or, in practice, a paralleled/stacked assembly of standard nanocrystalline C-cores or cut cores), consistent with a 250 kW MF transformer.

Step 3 — Core area and turns

Assuming a roughly square window/core split, $A_c \approx \sqrt{A_p} \approx 37\,\mathrm{cm^2}$. From Eq. (1):

$$N_1 = \frac{V_1}{K_f\, A_c\, B_m\, f} = \frac{800}{4 \times 37 \times 10^{-4} \times 1.0 \times 10{,}000} \approx 5.4 \;\Rightarrow\; N_1 = N_2 = 6 \text{ turns} \tag{12}$$

(rounding up gives margin against saturation; with $n = 1$, $N_2 = N_1$).

Step 4 — Conductor sizing

RMS current per winding (lossless approximation):

$$I_1 = I_2 = \frac{P}{V} = \frac{250{,}000}{800} = 312.5\,\mathrm{A} \tag{13}$$

Required copper cross-section at $J = 300\,\mathrm{A/cm^2}$:

$$A_{Cu} = \frac{I}{J} = \frac{312.5}{300} \approx 1.04\,\mathrm{cm^2} \tag{14}$$

With skin depth $\delta \approx 0.66\,\mathrm{mm}$ at 10 kHz, this area should be realized as a foil (or paralleled foils) of thickness $\leq 1.3\,\mathrm{mm}$ and sufficient width, or an equivalent litz bundle — foil/busbar is the practical choice at this current level and turn count.

Step 5 — Leakage inductance estimate

Assume a square-ish window of side $\approx 6.1\,\text{cm}$, so $h_w \approx 6\,\text{cm}$, and a mean turn length MLT $\approx$ 20 cm (typical for this core size). Each winding uses 6 foil turns of thickness 1.3 mm with 0.1 mm inter-turn insulation, giving a winding build-up:

$$b_1 = b_2 = 6 \times (1.3 + 0.1)\,\text{mm} = 8.4\,\text{mm} \tag{15}$$

With an inter-winding isolation gap $\delta = 2\,\text{mm}$, Eq. (6) gives:

$$L_{lk} \approx \frac{4\pi \times 10^{-7} \times 6^2 \times 0.20}{0.06}\left[\frac{0.0084 + 0.0084}{3} + 0.002\right] \approx 1.14\,\mu\text{H} \tag{16}$$

If the primary and secondary are interleaved into two sections each ($m = 2$), this drops to $L_{lk} \approx 0.29\,\mu\text{H}$. This value should be compared against the leakage/resonant inductance required by the DAB/PSFB power-transfer equation; if higher inductance is needed, an air gap or reduced interleaving can be used deliberately to tune L_{lk} upward.

Summary Table

Parameter	Value
Core material	Nanocrystalline (e.g. Vitroperm 500F)
B_m	1.0 T
Area product A_p	$\approx 1389\,\text{cm}^4$
$N_1 = N_2$	6 turns
$I_1 = I_2$ (RMS)	312.5 A
Winding type	Interleaved copper foil, $t \leq 1.3\,\text{mm}$
L_{lk} (non-interleaved / m=2)	$\approx 1.14\,\mu\text{H}$ / $0.29\,\mu\text{H}$

Step 6 — Loss estimate

Core loss. With $A_c \approx 37\,\text{cm}^2$ and window side $\approx 6.1\,\text{cm}$, $l_e \approx 2(\sqrt{37} + \sqrt{37}) \approx 24.4\,\text{cm}$, giving $V_e = A_c\, l_e \approx 925\,\text{cm}^3$. At $\rho_{mass} \approx 7.3\,\text{g/cm}^3$, core mass $\approx 6.75\,\text{kg}$. Assuming a typical nanocrystalline datasheet loss density at 10 kHz, 1.0 T of $P_v \approx 50\,\text{W/kg}$ (material- and temperature-dependent, to be confirmed against the actual manufacturer curve):

$$P_{core} \approx 50 \times 6.75 \approx 338\,\text{W} \tag{17}$$

Winding loss. R_{dc} per winding ($l = N \cdot \text{MLT} = 6 \times 0.20 = 1.2\,\text{m}$, $A_{Cu} = 1.04 \times 10^{-4}\,\text{m}^2$, $\rho_{Cu} \approx 2.2 \times 10^{-8}\,\Omega\,\text{m}$ at $\approx 100\,^\circ\text{C}$):

$$R_{dc} \approx \frac{2.2 \times 10^{-8} \times 1.2}{1.04 \times 10^{-4}} \approx 0.254\,\text{m}\Omega \quad \Rightarrow \quad P_{dc,\,1wdg} = I^2 R_{dc} \approx 24.8\,\text{W} \tag{18}$$

For both windings, $P_{dc} \approx 49.5\,\text{W}$. With $\Delta = t/\delta = 1.3/0.66 \approx 1.97$:

- Non-interleaved ($p = N = 6$): $F_R \approx 38.4 \Rightarrow P_{ac} \approx 1.9\,\text{kW}$ — unacceptably high, confirming interleaving is mandatory.
- Interleaved $m = 2$ ($p = 3$): $F_R \approx 10.2 \Rightarrow P_{ac} \approx 505\,\text{W}$.
- Fully interleaved ($p = 1$): $F_R \approx 1.86 \Rightarrow P_{ac} \approx 92\,\text{W}$.

Total loss (using the practical $m = 2$ interleaving case from the leakage calculation):

$$P_{tot} \approx P_{core} + P_{ac} \approx 338 + 505 \approx 843\,\text{W} \quad \left(\approx 0.34\%\text{ of } 250\,\text{kW}\right) \tag{19}$$

Notes

This first-pass sizing must be followed by: (i) FEA cross-check of the analytical leakage inductance estimate above (relevant for DAB/PSFB soft-switching range), (ii) thermal verification of core and winding losses against the chosen cooling method, and (iii) insulation coordination for the working voltage and dV/dt. The area-product method gives the correct order of magnitude but final core selection should match an actual manufacturer part (or stack thereof) with $A_p \geq$ the calculated value.

10 Alternative Design: Ferrite Core, Same Specification

For comparison, the same 800 V/800 V, 10 kHz, 250 kW transformer is now sized with a MnZn ferrite core.

Assumptions

- MnZn ferrite (e.g. 3C95/3C90-type material), $B_m = 0.3\,\mathrm{T}$ (derated from $B_{sat} \approx 0.4 - 0.45\,\mathrm{T}$ at 10 kHz and elevated temperature, to keep core losses bounded).
- Same $K_f = 4$, $K_u = 0.3$, $J = 300\,\mathrm{A/cm^2}$ as before, for direct comparison.

Step 1 — Area product

$$A_p = \frac{500{,}000 \times 10^4}{4 \times 0.3 \times 0.3 \times 10{,}000 \times 300} \approx 4630\,\mathrm{cm}^4 \tag{20}$$

This is roughly 3.3× larger than the nanocrystalline design, a direct consequence of the lower usable B_m — illustrating why ferrite becomes volumetrically penalized at high power and moderate (10 kHz) frequency, despite its lower specific core loss.

Step 2 — Core area and turns

$$A_c \approx \sqrt{A_p} \approx 68\,\mathrm{cm}^2 \tag{21}$$

$$N_1 = \frac{800}{4 \times 68 \times 10^{-4} \times 0.3 \times 10{,}000} \approx 9.8 \;\Rightarrow\; N_1 = N_2 = 10 \text{ turns} \tag{22}$$

Step 3 — Conductors

Current is unchanged ($I = 312.5\,\mathrm{A}$, $A_{Cu} \approx 1.04\,\mathrm{cm}^2$); skin depth at 10 kHz is also unchanged ($\delta \approx 0.66\,\mathrm{mm}$), so the same foil thickness limit ($\leq 1.3\,\mathrm{mm}$) applies, now over 10 turns instead of 6.

Step 4 — Leakage inductance

Window side $\approx 8.25\,\mathrm{cm} \Rightarrow h_w \approx 8\,\mathrm{cm}$; MLT $\approx 30\,\mathrm{cm}$ for the larger core. Winding build-up with 10 foil turns:

$$b_1 = b_2 = 10 \times (1.3 + 0.1)\,\mathrm{mm} = 14\,\mathrm{mm} \tag{23}$$

With the same $\delta = 2\,\mathrm{mm}$:

$$L_{lk} \approx \frac{4\pi \times 10^{-7} \times 10^2 \times 0.30}{0.08} \left[\frac{0.014 + 0.014}{3} + 0.002\right] \approx 5.3\,\mu\mathrm{H} \tag{24}$$

The higher turns count (N^2 scaling) and larger window both push leakage inductance up by roughly 4.7× relative to the nanocrystalline design — a relevant trade-off if a specific L_{lk} is targeted for soft-switching.

Comparison Summary

Parameter	Nanocrystalline	Ferrite (MnZn)
B_m	1.0 T	0.3 T
A_p	$\approx 1389\,\text{cm}^4$	$\approx 4630\,\text{cm}^4$
$N_1 = N_2$	6	10
L_{lk} (non-interleaved)	$\approx 1.14\,\mu\text{H}$	$\approx 5.3\,\mu\text{H}$
Relative core volume	baseline	$\approx 3.3\times$

At this power and frequency, nanocrystalline gives a markedly more compact design with lower leakage; ferrite remains attractive mainly where its higher achievable switching frequency (tens to hundreds of kHz) can be exploited, which is not the case at 10 kHz.

Ferrite Variant with $N = 6$ (Re-proportioned Core)

The area product $A_p = A_c \cdot A_w$ fixes only the *product* of core cross-section and window area, not their individual split. It is therefore possible to keep the ferrite design at $N = 6$ turns (matching the nanocrystalline case) simply by choosing a "short and fat" core geometry — larger A_c, smaller A_w — instead of the roughly square proportions assumed above.

Solving Eq. (1) for the core area required at $N = 6$:

$$A_c = \frac{V_1}{K_f\, N\, B_m\, f} = \frac{800}{4 \times 6 \times 0.3 \times 10{,}000} \approx 111\,\text{cm}^2 \tag{25}$$

$$A_w = \frac{A_p}{A_c} = \frac{4630}{111} \approx 41.7\,\text{cm}^2 \tag{26}$$

This A_c is $\approx 1.6\times$ larger than the nanocrystalline core's, and the window is correspondingly smaller — but the total A_p, and hence the overall core volume/mass, is unchanged from the square-proportioned ferrite design: only the *aspect ratio* of the core has changed.

Leakage inductance. With $N = 6$ foil turns (same build-up as the nanocrystalline example, $b_1 = b_2 = 8.4\,\text{mm}$) and a smaller, taller window ($h_w \approx 6.5\,\text{cm}$, with a wider leg giving MLT $\approx 38\,\text{cm}$):

$$L_{lk} \approx \frac{4\pi \times 10^{-7} \times 6^2 \times 0.38}{0.065} \left[\frac{0.0084 + 0.0084}{3} + 0.002\right] \approx 2.0\,\mu\text{H} \tag{27}$$

This sits between the nanocrystalline (1.14 µH) and the square-proportioned ferrite (5.3 µH) cases: at equal turns count, the wider core leg increases MLT relative to h_w, partially offsetting the leakage reduction gained from fewer turns.

Trade-off. Reducing N via a fatter core does *not* reduce ferrite's fundamental disadvantage at this operating point — the core remains $\approx 3.3\times$ larger in volume than nanocrystalline because B_m is lower. What changes is only the turns count and the leakage inductance value, which can be useful if a specific winding configuration (e.g. matching busbar terminations) is desired, or if a target L_{lk} is being tuned via geometry rather than turns count.

Loss estimate for this variant. *Core loss.* With $A_c \approx 111\,\text{cm}^2$ and window side $\approx 6.46\,\text{cm}$, leg side $\approx \sqrt{111} \approx 10.5\,\text{cm}$, so $l_e \approx 2(\sqrt{111} + \sqrt{41.7}) \approx 34\,\text{cm}$, giving $V_e = A_c\, l_e \approx 3774\,\text{cm}^3$. Assuming a typical MnZn ferrite (3C95-class) loss density at 10 kHz, 0.3 T of $P_v \approx 30\,\text{mW/cm}^3$ (datasheet-dependent, generally low at this B/f combination):

$$P_{core} \approx 30 \times 10^{-3} \times 3774 \approx 113\,\text{W} \tag{28}$$

Despite the much larger volume, ferrite's lower operating B_m keeps its core loss *below* the nanocrystalline value — the trade-off shows up almost entirely in the windings.

Winding loss. $l = N \cdot \mathrm{MLT} = 6 \times 0.38 = 2.28\,\mathrm{m}$ per winding:

$$R_{dc} \approx \frac{2.2 \times 10^{-8} \times 2.28}{1.04 \times 10^{-4}} \approx 0.482\,\mathrm{m\Omega} \quad \Rightarrow \quad P_{dc,\,1wdg} \approx 47.1\,\mathrm{W} \tag{29}$$

For both windings, $P_{dc} \approx 94.2\,\mathrm{W}$. Δ is unchanged (≈ 1.97, same foil thickness and frequency), so F_R values are the same as before ($p = 6$ turns/layers):

- Non-interleaved: $F_R \approx 38.4 \Rightarrow P_{ac} \approx 3.6\,\mathrm{kW}$.
- Interleaved $m = 2$ ($p = 3$): $F_R \approx 10.2 \Rightarrow P_{ac} \approx 961\,\mathrm{W}$.
- Fully interleaved ($p = 1$): $F_R \approx 1.86 \Rightarrow P_{ac} \approx 175\,\mathrm{W}$.

Total loss ($m = 2$ interleaving, for consistency with the nanocrystalline case):

$$P_{tot} \approx 113 + 961 \approx 1074\,\mathrm{W} \quad \big(\approx 0.43\%\text{ of }250\,\mathrm{kW}\big) \tag{30}$$

Loss Comparison: Nanocrystalline vs. Ferrite (N=6 variant)

	Nanocrystalline	Ferrite (N=6 variant)
P_{core}	338 W	113 W
P_{dc} (both windings)	49.5 W	94.2 W
P_{ac} (m=2 interleaving)	505 W	961 W
P_{tot}	$\approx$ 843 W	$\approx$ 1074 W
Loss / rated power	$\approx 0.34\%$	$\approx 0.43\%$

The ferrite variant has *lower* core loss (lower B_m more than offsets the larger volume) but *higher* copper loss, driven by the longer MLT of the wider core leg at equal turns count. The net result is a $\approx 27\%$ higher total loss for the ferrite (N=6) solution relative to nanocrystalline, at the same interleaving strategy. All values are first-pass estimates and should be refined with actual datasheet loss curves and a 2D/3D field simulation of the specific winding arrangement.

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